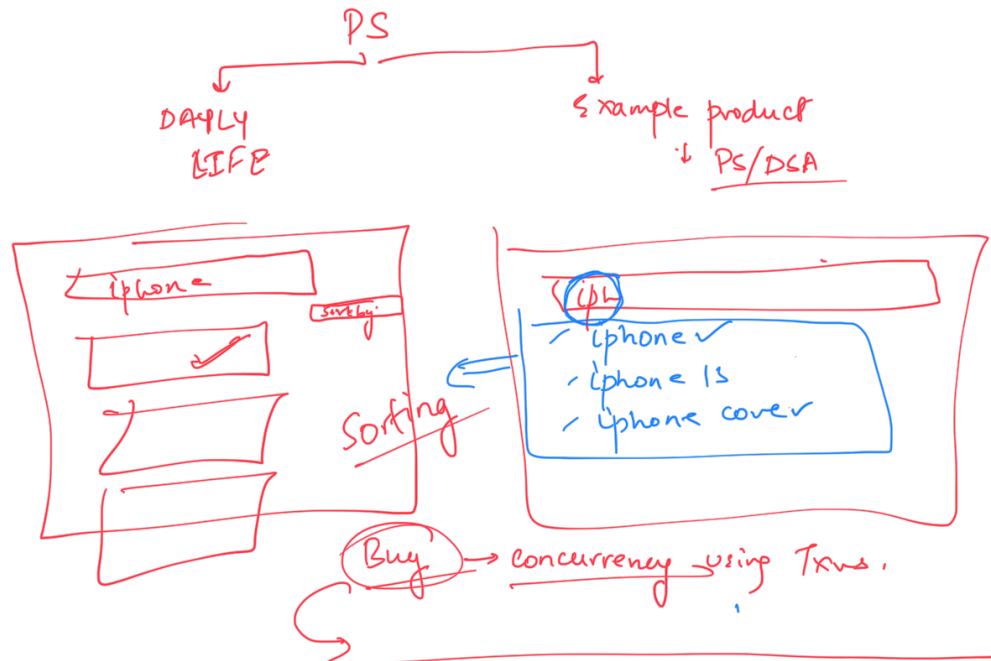


INTRO TO PROBLEM SOLVING



PS: ① Observation | ② log | ③ Basic intro & time comp

$$S = \underline{1} + \overset{+1}{\textcircled{2}} + \textcircled{3} + 4 + \dots \dots \dots \textcircled{98} + \overset{-1}{\textcircled{99}} + \underline{100}$$

$$\left[\frac{100}{2} \times 101 \right] \quad \checkmark \frac{10}{2} = 101$$

$$= 5050$$

$$\textcircled{1} + \textcircled{2} + \textcircled{3} + \textcircled{4} + \textcircled{5} + \textcircled{6}$$

every pair = 2 numbers.

how many pairs will cover
6 numbers = $\frac{6}{2} = 3$

$$100 \text{ num} = \frac{100}{2} = 50$$

$$\textcircled{1} + \underline{2} + \underline{3} + \underline{4} + \boxed{5} + \underline{6} + \underline{7} + \underline{8} + \textcircled{9}$$

... (9-1) ... 10)

$$10 * \frac{4}{2} + 5$$

$$= 10 * 4 + 5 = \underline{45}$$

10
10
10

$$1 + 2 + 3 + \dots + N$$

$$= \left[\frac{N}{2} * (N+1) \right]$$

N is odd
(N-1) is even

$$1 + 2 + \dots + (N-1) + N$$

$$\frac{(N-1)}{2} * N + N = N \left(\frac{N-1}{2} + 1 \right)$$

$$= N \left(\frac{N-1+2}{2} \right) = N \left(\frac{N+1}{2} \right)$$

Prime numbers

$$\boxed{100} = N$$

$$\boxed{10} \rightarrow d$$

$$\frac{100}{10} = \underline{\underline{10}}$$

divisor or a factor

$$(N \% d == 0)$$

$$\underline{6} \quad \underline{8}$$

$$100 \rightarrow 1, 2, 4, 5, 10, \dots$$

$$\underline{15} \rightarrow 1, \underline{3}, \underline{5}, 15 \rightarrow 15 \text{ is not prime}$$

$$2 \text{ factors} \rightarrow 1 \text{ and itself}$$

$$23 \rightarrow 1, 23$$

Time complexity ↗

Check Prime (N) :

Count-factors = 0

i → 1 to N :

if (N % i == 0) :

Count-factors += 1

if Count-factors == 2 :

$\downarrow$ return true
 $\downarrow$ else return false

10 seconds Assumption: $10^8 \text{ ops} \rightarrow 1 \text{ second}$ Atk at
 10^9 $\rightarrow \{1 \text{ to } 10^9\}$ 10^9 ops.

$$\frac{10^8 \text{ ops}}{1 \text{ op}} \rightarrow \frac{1 \text{ second}}{10^8} \text{ seconds}$$

$$\frac{10^9 \text{ ops}}{10^8 \text{ ops}} \rightarrow \frac{1}{10} * 10^9 \text{ seconds}$$

$$= \frac{10 * 10 * 10 * \dots * 10}{10 * 10 * 10 * \dots * 10} \text{ (9 times / 8 times)}$$

$\checkmark$ $a^b = a * a * a * a * \dots \text{ b times} = 10$
 10 seconds.

$$\frac{5 \text{ ops}}{1 \text{ op}} \rightarrow \frac{1 \text{ second}}{5} \text{ seconds.}$$

$$\frac{10^8 \text{ ops}}{1 \text{ ops}} \rightarrow \frac{1 \text{ second}}{10^8} \text{ seconds}$$

$$\frac{10^9 \text{ ops}}{10^8 \text{ ops}} \rightarrow \frac{1}{10} \text{ seconds} * 10^9$$

$$\rightarrow \frac{10}{10^8} = \frac{10 * 10 * 10 * \dots * 10}{10 * 10 * 10 * \dots * 10} \text{ (9 times / 8 times)}$$

$$= 10$$

10^{18} $\rightarrow$ prime or not

$$10^8 \text{ ops} \rightarrow 1 \text{ sec}$$

$$1 \text{ op} \rightarrow \frac{1}{10^8} \text{ sec.}$$

1 day → 86400 seconds

365 days → 31 million

~~10 years~~

$$10^{18} \text{ ops} = \frac{10^8}{10} * 10^{18}$$

$$= 10 * 10 * 10 * \dots * 10 \text{ times}$$

$$= 10 \text{ billion seconds.}$$

~~to be 10,000 million sec~~
31 million sec

~ 300 years.

1 — N
 2 or bigger
 ↓
 1 and the number
 will be factor

8 → 1, 8

2, 3, 4, 5, 6, 7 18
10

check-prime (N):
 if $N < 2$:
 return false

if $N \leq \frac{N}{2}$:
 if $N \% i == 0$:
 return false
 return true
 prime

$$10^{18} \rightarrow 10^{18}/2$$

300
 ↓
 150 years

$N = a * b$ $a \leq b$ $\left(\frac{N}{a}\right)$

8 ← 2 (4) $8 = 2 * 4$ $a \geq 2$ $a \leq N$ $d \leftarrow N/d$ $N = d * \left(\frac{N}{d}\right)$

N → any one factor
 apart from 1 and N

100 = 2 * 50
4 * 25

$N = \underline{a} * \underline{b}$

what is largest value for

$\frac{N}{b=1}$
 $a \leq b$
 $(a \leq b)$

$$5 * 20 \quad \checkmark$$

$$10 * 10 \quad \checkmark$$

$$a \leq b$$

$$\checkmark a = \frac{N}{2}, b = 2 \quad a \leq b \quad \checkmark$$

$$N = \uparrow a * b \downarrow \quad a \leq b$$

$$a == b$$

$$a * a = N$$

$$a = \underline{\underline{\text{sqrt}(N)}}$$

$$24 \begin{cases} \rightarrow 2 * 12 \\ \rightarrow 3 * 8 \\ \rightarrow 4 * 6 \end{cases}$$

$$< \text{sqrt}(24) = 4.8989...$$

$$N = \uparrow a * b \downarrow$$

a is smaller than or equal to b

$$N = \underline{\underline{a * b}}$$

$$\downarrow$$

$$\leq \text{sqrt}(N)$$

~~9 to~~

99

9...

$$\underline{\underline{2}} \quad \underline{\underline{9}}$$

$$99 \rightarrow 2 \times$$

$$99 \rightarrow 3 \checkmark$$

$$\underline{\underline{10^{18}}} \rightarrow 10^9$$

$$i \rightarrow 2 \text{ to } 10^9 \times$$

$$\boxed{i \leq \text{sqrt}(N)}$$

$$i * i \leq N$$

check Prime (N) :

if $N < 2$:

return false

for ($i = 1$; $i * i \leq N$; $i++$)

if $N \% i == 0$:

return false

return true

$$\underline{\underline{10^{18}}}$$

$$\text{sqrt}(10^{18}) = \underline{\underline{10^9}}$$

$$\underline{\underline{i = \sqrt{N}}}$$

$$\underline{\underline{i^2 = N}}$$

$$\underline{\underline{10^9 \text{ ops}}}$$

$$1 \text{ ops} \rightarrow \frac{1}{10^8} \text{ second}$$

$$10^7 \text{ ops} \rightarrow \frac{10^7}{10^8} \text{ sec} = 10^{-1}$$

N → prime
N → not prime

$$N = a * b \quad a \leq b$$

Sqrt

$$N \rightarrow N/2 \rightarrow N/4 \rightarrow N/8 \dots \textcircled{1}$$

$$5/2$$

②

~~2~~

↑ integer divisions

N = 10

10/2 = 5

5/2 = 2

2/2 = 1

10x times

Ans = 3

2x time

N = 100

100/2 = 50

50/2 = 25

25/2 = 12

12/2 = 6

6/2 = 3

3/2 = 1

Ans = 6

1

N → N/2

Step 1

N/2

Step 2

N/2

Step 3

N/2

Step 4

N/2

i-th step → $\frac{N}{2^i} = 1$

$$N = 2^{ni}$$

$$\sqrt{x} = N$$

$$\frac{N}{2^i} = N$$

x

$$\log 2^i = \log N$$

$$\log 2 = (X)$$

$$i \cdot \log_2 = \log_2 N$$

$$i = \log_2 N$$

$$\log_{\text{base}} a = x$$

$$\text{base}^x = a$$

$$\log_{\text{base}} a = \frac{\log a}{\log \text{base}}$$

$$\log_{10} a$$

$$\log_{10} \text{base}$$

$$2^i = N$$

$$\log_2 2^i = \log_2 N$$

$$i = \log_2 N$$

$$\log_{\text{base}} N$$

$$= \frac{\log N}{\log \text{base}}$$

$$\begin{matrix} 2 & 3 \\ & 5 \\ & 6 \end{matrix}$$

$$2^i = N$$

$$8 \rightarrow \log_2 8 = 3$$

$$16 \rightarrow \log_2 16 = 4$$

$$1000 \rightarrow 10000$$

$$i = 10 \rightarrow 13$$

$$\text{Sqrt}(N)$$

$$N = 10^9$$

$$\text{ans} > 1000$$

$$\sqrt{N} \leftarrow \text{Ans}$$

$$1000$$

$$1000^2 = 10^6 < N$$

$$i$$

$$i^2$$

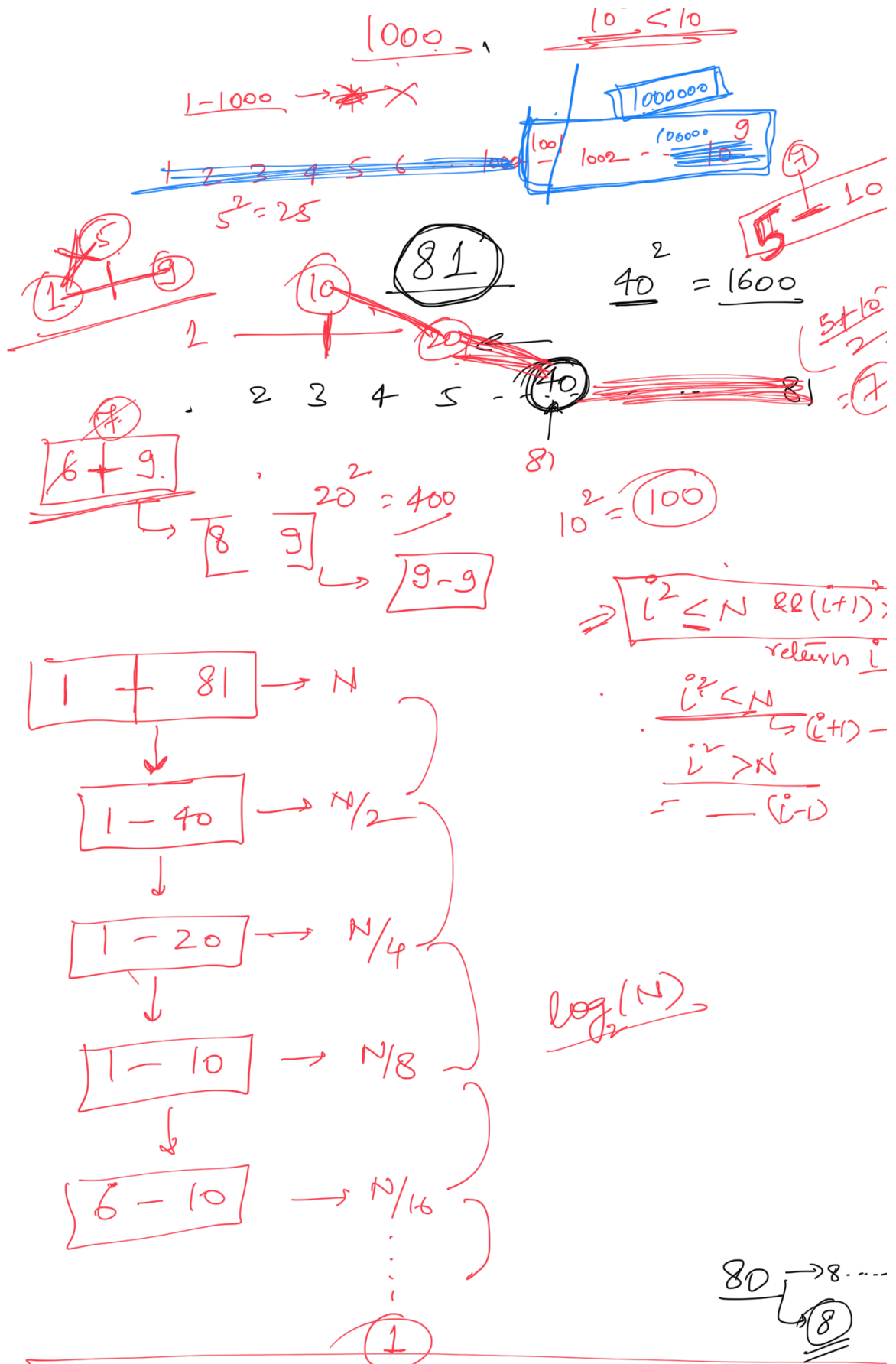
$$< N$$

$$\text{ans} > i$$

$$i^2 > N$$

$$\text{ans} < i$$

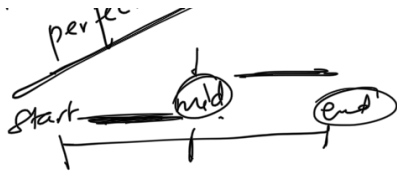
$$10^9$$



sqrt(N) :

start = 1, end = N
while (start != end)

N is a perfect square



$\log N$

while (start <= end) {

$$\text{mid} = \frac{\text{start} + \text{end}}{2}$$

if ($\text{mid}^2 \leq N$ && $(\text{mid}+1)^2 > N$)

return mid

if $\text{mid} * \text{mid} > N$:

end = mid - 1

else

start = mid + 1

N

Q & A

$$i \leq \sqrt{N}$$

↓

$$i * i \leq N$$

Sqrt(N)

i → 2 to $\sqrt{N}$:

if $N \% i == 0$:

return false

return true

$$N = a * b$$

$$a \leq \sqrt{N}$$

for (i = 2; $i \leq \sqrt{N}$; i++)

return false

$$i \leq \sqrt{N}$$

i = 2; $i \leq \sqrt{N}$; i++

$$i \leq \sqrt{N}$$

$$i * i \leq N$$

$$i * i \leq \sqrt{N} * \sqrt{N}$$

$$i^2 \leq N$$